

# **TAGORE COLLEGE OF ARTS AND SCIENCE**

**(Affiliated to University of Madras)  
Chrompet, Chennai-44**



## **DEPARTMENT OF COMPUTER SCIENCE**

### **RELATIONAL DATABASE MANAGEMENT SYSTEM**

**NAME :**

**CLASS :**

**REG NO :**

# TAGORE COLLEGE OF ARTS AND SCIENCE

(Affiliated to the University of Madras)

Chrompet, Chennai – 600 044



## DEPARTMENT OF COMPUTER SCIENCE

Certified to be the Bonafide Record of Practical work done by Mr. /Miss.  
..... Reg. No..... of Third Year B.Sc., in  
the Department of Computer Science, Shift-I, Tagore College of Arts & Science,  
Chrompet, Chennai – 600 044 during the year 2026-2027

STAFF IN-CHARGE

HEAD OF THE DEPARTMENT

Submitted for B.Sc., Degree Practical Examination held on .....  
at the Department of Computer Science, Shift-I, Tagore College of Arts and  
Science, Chrompet, Chennai –44.

Date :

Examiners

1.

2.

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## **1)DDL COMMENTS WITH CONSTRAINTS**

### **DDL commands:**

- \* CREATE
- \* ALTER
- \* DROP
- \* RENAME
- \* TRUNCATE

### **Queries:**

#### **CREATE TABLE**

Q1 → create table student(stid number, sname varchar(20), score integer);

Q2 → desc student;

Q3 → create table stu(sid number primary key, sname varchar(20) not null, email varchar(30) unique, age decimal(10,2) check (age>=18));

Q4 → desc stu;

#### **ALTER TABLE**

Q5 → alter table student drop column score;

Q6 → alter table student add gender varchar(5);

Q7 → alter table stu modify age number;

#### **RENAME TABLE**

Q8 → rename student to stu10;

Q9 → desc stu10;

Q10 → alter table stu rename column email to emailid;

Q11 → desc stu;

### **TRUNCATE AND DROP**

Q12 → truncate table stu;

Q13 → drop table stu;

Q14 → desc stu;

## OUTPUT:

### CREATE TABLE

O/P → Table created.

| Table          | Column        | Data Type | Length | Precision | Scale | Primary Key | Nullable | Default | Comment |
|----------------|---------------|-----------|--------|-----------|-------|-------------|----------|---------|---------|
| <u>STUDENT</u> | <u>STID</u>   | Number    | -      | -         | -     | -           | ✓        | -       | -       |
|                | <u>STNAME</u> | Varchar2  | 20     | -         | -     | -           | ✓        | -       | -       |
|                | <u>SCORE</u>  | Number    | -      | -         | 0     | -           | ✓        | -       | -       |

O/P →

O/P → Table created.

O/P →

| Table      | Column       | Data Type | Length | Precision | Scale | Primary Key | Nullable | Default | Comment |
|------------|--------------|-----------|--------|-----------|-------|-------------|----------|---------|---------|
| <u>STU</u> | <u>SID</u>   | Number    | -      | -         | -     | 1           | -        | -       | -       |
|            | <u>SNAME</u> | Varchar2  | 20     | -         | -     | -           | -        | -       | -       |
|            | <u>EMAIL</u> | Varchar2  | 30     | -         | -     | -           | ✓        | -       | -       |
|            | <u>AGE</u>   | Number    | -      | 10        | 2     | -           | ✓        | -       | -       |

### ALTER TABLE

O/P →

| Table          | Column        | Data Type | Length | Precision | Scale | Primary Key | Nullable | Default | Comment |
|----------------|---------------|-----------|--------|-----------|-------|-------------|----------|---------|---------|
| <u>STUDENT</u> | <u>STID</u>   | Number    | -      | -         | -     | -           | ✓        | -       | -       |
|                | <u>STNAME</u> | Varchar2  | 20     | -         | -     | -           | ✓        | -       | -       |

O/P → Table altered.

O/P → Table altered.

| Table      | Column       | Data Type | Length | Precision | Scale | Primary Key | Nullable | Default | Comment |
|------------|--------------|-----------|--------|-----------|-------|-------------|----------|---------|---------|
| <u>STU</u> | <u>SID</u>   | Number    | -      | -         | -     | 1           | -        | -       | -       |
|            | <u>SNAME</u> | Varchar2  | 20     | -         | -     | -           | -        | -       | -       |
|            | <u>EMAIL</u> | Varchar2  | 30     | -         | -     | -           | ✓        | -       | -       |
|            | <u>AGE</u>   | Number    | -      | -         | -     | -           | ✓        | -       | -       |

### RENAME TABLE

O/P → Statement processed.

O/P →

| Table        | Column        | Data Type | Length | Precision | Scale | Primary Key | Nullable | Default | Comment |
|--------------|---------------|-----------|--------|-----------|-------|-------------|----------|---------|---------|
| <u>STU10</u> | <u>STID</u>   | Number    | -      | -         | -     | -           | ✓        | -       | -       |
|              | <u>STNAME</u> | Varchar2  | 20     | -         | -     | -           | ✓        | -       | -       |

|  |                        |          |   |   |   |   |   |   |   |
|--|------------------------|----------|---|---|---|---|---|---|---|
|  | <a href="#">GENDER</a> | Varchar2 | 5 | - | - | - | ✓ | - | - |
|--|------------------------|----------|---|---|---|---|---|---|---|

O/P →

| Table               | Column                | Data Type | Length | Precision | Scale | Primary Key | Nullable | Default | Comment |
|---------------------|-----------------------|-----------|--------|-----------|-------|-------------|----------|---------|---------|
| <a href="#">STU</a> | <a href="#">SID</a>   | Number    | -      | -         | -     | 1           | -        | -       | -       |
|                     | <a href="#">SNAME</a> | Varchar2  | 20     | -         | -     | -           | -        | -       | -       |
|                     | <a href="#">EMAIL</a> | Varchar2  | 30     | -         | -     | -           | ✓        | -       | -       |
|                     | <a href="#">AGE</a>   | Number    | -      | -         | -     | -           | ✓        | -       | -       |

## **TRUNCATE AND DROP**

O/P → Table truncated.

O/P → Table dropped.

O/P → Object to be described could not be found.

## **2)DML COMMENTS WITH CONSTRAINTS**

### **DML commands:**

- \* INSERT
- \* UPDATE
- \* DELETE
- \* SELECT

### **Queries:**

#### **INSERT QUERY**

Q1 → insert into stu10(stdid,sname,gender) values (1001,'aaa','f');

Q2 → insert into stu10(stdid,sname,gender) values (1002,'bbb','f');

Q3 → insert into stu10(stdid,sname,gender) values (1003,'cccc','f');

Q4 → insert into stu10(stdid,sname,gender) values (1004,'ddd','m');

Q5 → insert into stu10(stdid,sname,gender) values (1005,'EEE','M');

#### **SELECT QUERY**

Q6 → select \* from stu10;

Q7 → select sname from stu10;

Q8 → select \* from stu10 where gender='f';

#### **UPDATE QUERY**

Q9 → update stu10 set sname='anu' where stdid=1001;

Q10 → update stu10 set stname='bala', gender='m' where stid=1002;

Q11 → select \* from stu10;

### **DELETE QUERY**

Q12 → delete from stu10 where stid=1003;

Q13 → select \* from stu10;

Q14 → delete from stu10;

Q15 → select \* from stu10;

### OUTPUT:

### INSERT QUERY

O/P → 1 row(s) inserted.

### SELECT QUERY

O/P

→

| STID | STNAME | GENDER |
|------|--------|--------|
| 1001 | aaa    | f      |
| 1002 | bbb    | f      |
| 1003 | cccc   | f      |
| 1004 | ddd    | m      |
| 1005 | EEE    | M      |

O/P →

| STNAME |
|--------|
| aaa    |
| bbb    |
| cccc   |
| ddd    |
| EEE    |

O/P →

| STID | STNAME | GENDER |
|------|--------|--------|
| 1001 | aaa    | f      |

|      |      |   |
|------|------|---|
| 1002 | bbb  | f |
| 1003 | cccc | f |

### **UPDATE QUERY**

O/P → 1 row(s) updated.

O/P → 1 row(s) updated.

O/P →

| STID | STNAME | GENDER |
|------|--------|--------|
| 1001 | anu    | f      |
| 1002 | bala   | m      |
| 1003 | cccc   | f      |
| 1004 | ddd    | m      |
| 1005 | EEE    | M      |

### **DELETE QUERY**

O/P → 1 row(s) deleted.

O/P →

| STID | STNAME | GENDER |
|------|--------|--------|
| 1001 | anu    | f      |
| 1002 | bala   | m      |
| 1004 | ddd    | m      |

|      |     |   |
|------|-----|---|
| 1005 | EEE | M |
|------|-----|---|

O/P → 4 row(s) deleted.

O/P → no data found

### **3)QUERIES, SUBQUERIES AND AGGREGATE FUNCTION**

#### **Queries:**

#### **BASIC QUERY**

Q1 → create table empl(eid number primary key, ename varchar(20), deptid number, deptname varchar(20),salary number);

Q2 → insert into empl(eid,ename,deptid,deptname,salary) values (1201,'Arun',104,'CS',50000);

Q3 → insert into empl(eid,ename,deptid,deptname,salary) values (1202,'Bala',104,'CS',60000);

Q4 →insert into empl(eid,ename,deptid,deptname,salary) values (1203,'Chandru',105,'COMMERCE',40000);

Q5 →insert into empl(eid,ename,deptid,deptname,salary) values (1204,'Dilip',104,'CS',45000);

Q6 →insert into empl(eid,ename,deptid,deptname,salary) values (1205,'Dhanu',106,'TAMIL',30000);

Q7 →insert into empl(eid,ename,deptid,deptname,salary) values (1206,'Guru',106,'TAMIL',58000);

Q8 →insert into empl(eid,ename,deptid,deptname,salary) values (1207,'Hari',105,'COMMERCE',38000);

## **SELECT QUERY**

Q9 → select \* from empl;

Q10 → select eid,ename,salary from empl;

Q11 → select \* from empl where salary>=50000;

## **AGGREGATE FUNCTION**

Q12 → select COUNT(\*) as totalemp from empl;

Q13 → select SUM(salary) as totalsal from empl;

Q14 → select AVG(salary) as totalsal from empl;

Q15 →select MAX(salary) as maxsalary from empl;

Q16 →select MIN(salary) as minsalary from empl;

Q17 →select eid,salary from empl order by salary desc;

Q18 →select deptid, sum(salary)as sal from empl group by deptid;

## **SUBQUERIES**

Q19 → select \* from empl where salary > (select avg(salary) from empl);

Q20 →create table dept(deptid number,deptname varchar(20));

Q21 →insert into dept(deptid,deptname) values (104,'CS');

Q22 →insert into dept(deptid,deptname) values (105,'COMMERCE');

Q23 →insert into dept(deptid,deptname) values (106,'TAMIL');

Q24 →insert into dept(deptid,deptname) values (107,'ENGLISH');

Q25 →select \* from dept;

--- List department wise employee count:

Q26 →select d.deptname, e.empcount from dept d JOIN (select deptid, count(\*) as empcount  
from empl group by deptid) e on d.deptid = e.deptid;

## **JOIN QUERIES**

Q27 →select e.eid, e.ename, e.salary, d.deptid, d.deptname from empl e join dept d on  
e.deptid = d.deptid;

Q28 → select e.eid, e.ename, e.salary, d.deptid, d.deptname from empl e LEFT JOIN dept d  
on e.deptid = d.deptid;

Q29 → select e.eid, e.ename, e.salary, d.deptid, d.deptname from empl e RIGHT JOIN dept d  
on e.deptid = d.deptid;

Q30 → select e.eid, e.ename, e.salary, d.deptid, d.deptname from empl e CROSS JOIN dept  
d;

## **OUTPUT:**

### **BASIC QUERY**

O/P → Table created.

### **SELECT QUERY**

O/P

→

| EID  | ENAME   | DEPTID | DEPTNAME | SALARY |
|------|---------|--------|----------|--------|
| 1201 | Arun    | 104    | CS       | 50000  |
| 1202 | Bala    | 104    | CS       | 60000  |
| 1203 | Chandru | 105    | COMMERCE | 40000  |
| 1204 | Dilip   | 104    | CS       | 45000  |
| 1205 | Dhanu   | 106    | TAMIL    | 30000  |
| 1206 | Guru    | 106    | TAMIL    | 58000  |
| 1207 | Hari    | 105    | COMMERCE | 38000  |

O/P →

| EID  | ENAME   | SALARY |
|------|---------|--------|
| 1201 | Arun    | 50000  |
| 1202 | Bala    | 60000  |
| 1203 | Chandru | 40000  |
| 1204 | Dilip   | 45000  |
| 1205 | Dhanu   | 30000  |
| 1206 | Guru    | 58000  |
| 1207 | Hari    | 38000  |

O/P →

| EID  | ENAME | DEPTID | DEPTNAME | SALARY |
|------|-------|--------|----------|--------|
| 1201 | Arun  | 104    | CS       | 50000  |
| 1202 | Bala  | 104    | CS       | 60000  |
| 1206 | Guru  | 106    | TAMIL    | 58000  |

### AGGREGATE FUNCTION

O/P →

| TOTALEMP |
|----------|
| 7        |

O/P →

| TOTALSAL |
|----------|
| 321000   |

O/P →

| TOTALSAL                           |
|------------------------------------|
| 45857.1428571428571428571428571429 |

O/P →

| MAXSALARY |
|-----------|
| 60000     |

O/P

→

| <b>MINSALARY</b> |
|------------------|
| 30000            |

O/P

→

| <b>EID</b> | <b>SALARY</b> |
|------------|---------------|
| 1202       | 60000         |
| 1206       | 58000         |
| 1201       | 50000         |
| 1204       | 45000         |
| 1203       | 40000         |
| 1207       | 38000         |
| 1205       | 30000         |

O/P →

| <b>DEPTID</b> | <b>SAL</b> |
|---------------|------------|
| 104           | 155000     |
| 105           | 78000      |
| 106           | 88000      |

## **SUBQUERIES**

O/P →

| <b>EID</b> | <b>ENAME</b> | <b>DEPTID</b> | <b>DEPTNAME</b> | <b>SALARY</b> |
|------------|--------------|---------------|-----------------|---------------|
|------------|--------------|---------------|-----------------|---------------|

|      |      |     |       |       |
|------|------|-----|-------|-------|
| 1201 | Arun | 104 | CS    | 50000 |
| 1202 | Bala | 104 | CS    | 60000 |
| 1206 | Guru | 106 | TAMIL | 58000 |

O/P →

| DEPTID | DEPTNAME |
|--------|----------|
| 104    | CS       |
| 105    | COMMERCE |
| 106    | TAMIL    |
| 107    | ENGLISH  |

O/P →

| DEPTNAME | EMPCOUNT |
|----------|----------|
| COMMERCE | 2        |
| CS       | 3        |
| TAMIL    | 2        |

### **JOIN QUERIES**

O/P →

| EID  | ENAME   | SALARY | DEPTID | DEPTNAME |
|------|---------|--------|--------|----------|
| 1201 | Arun    | 50000  | 104    | CS       |
| 1202 | Bala    | 60000  | 104    | CS       |
| 1203 | Chandru | 40000  | 105    | COMMERCE |

|      |       |       |     |          |
|------|-------|-------|-----|----------|
| 1204 | Dilip | 45000 | 104 | CS       |
| 1205 | Dhanu | 30000 | 106 | TAMIL    |
| 1206 | Guru  | 58000 | 106 | TAMIL    |
| 1207 | Hari  | 38000 | 105 | COMMERCE |

O/P →

| EID  | ENAME   | SALARY | DEPTID | DEPTNAME |
|------|---------|--------|--------|----------|
| 1204 | Dilip   | 45000  | 104    | CS       |
| 1202 | Bala    | 60000  | 104    | CS       |
| 1201 | Arun    | 50000  | 104    | CS       |
| 1207 | Hari    | 38000  | 105    | COMMERCE |
| 1203 | Chandru | 40000  | 105    | COMMERCE |
| 1206 | Guru    | 58000  | 106    | TAMIL    |
| 1205 | Dhanu   | 30000  | 106    | TAMIL    |

O/P →

| EID  | ENAME   | SALARY | DEPTID | DEPTNAME |
|------|---------|--------|--------|----------|
| 1201 | Arun    | 50000  | 104    | CS       |
| 1202 | Bala    | 60000  | 104    | CS       |
| 1203 | Chandru | 40000  | 105    | COMMERCE |
| 1204 | Dilip   | 45000  | 104    | CS       |
| 1205 | Dhanu   | 30000  | 106    | TAMIL    |
| 1206 | Guru    | 58000  | 106    | TAMIL    |

|      |      |       |     |          |
|------|------|-------|-----|----------|
| 1207 | Hari | 38000 | 105 | COMMERCE |
| -    | -    | -     | 107 | ENGLISH  |

O/P →

| EID  | ENAME   | SALARY | DEPTID | DEPTNAME |
|------|---------|--------|--------|----------|
| 1201 | Arun    | 50000  | 104    | CS       |
| 1202 | Bala    | 60000  | 104    | CS       |
| 1203 | Chandru | 40000  | 104    | CS       |
| 1204 | Dilip   | 45000  | 104    | CS       |
| 1205 | Dhanu   | 30000  | 104    | CS       |
| 1206 | Guru    | 58000  | 104    | CS       |
| 1207 | Hari    | 38000  | 104    | CS       |
| 1201 | Arun    | 50000  | 105    | COMMERCE |
| 1202 | Bala    | 60000  | 105    | COMMERCE |
| 1203 | Chandru | 40000  | 105    | COMMERCE |

#### **4)PL/SQL – EXCEPTION HANDLING**

##### **Queries:**

##### **Predefined Exceptions**

Q1 →

```
DECLARE

num1 NUMBER := 10;
num2 NUMBER := 0;
result NUMBER;

BEGIN

result := num1 / num2;

DBMS_OUTPUT.PUT_LINE('Result: ' || result);

EXCEPTION

WHEN ZERO_DIVIDE THEN

DBMS_OUTPUT.PUT_LINE('Cannot divide by zero.');
```

END;

##### **User-Defined Exceptions**

Q2 →

```
DECLARE

insufficient_balance EXCEPTION;

balance NUMBER := 500;

BEGIN

IF balance < 1000 THEN

RAISE insufficient_balance;

END IF;

EXCEPTION

WHEN insufficient_balance THEN
```

```
DBMS_OUTPUT.PUT_LINE('Balance is too low.');
```

END;

### **Named System Exceptions**

Q3 →

```
DECLARE

ename emp.name%TYPE;

BEGIN

SELECT name into ename

FROM emp

WHERE eid = 1001; -- assuming invalid ID

DBMS_OUTPUT.PUT_LINE('Employee Name: ' || ename);

EXCEPTION

WHEN NO_DATA_FOUND THEN

DBMS_OUTPUT.PUT_LINE('No employee found with that ID.');
```

END;

### **Unhandled (OTHERS) Exception**

Q4 →

```
BEGIN

EXECUTE IMMEDIATE 'DROP TABLE nonexistent_table';

EXCEPTION

WHEN OTHERS THEN

DBMS_OUTPUT.PUT_LINE('Some unexpected error occurred: ' || SQLERRM);

END;
```

## **OUTPUT:**

### **Predefined Exceptions**

O/P

→

Cannot divide by zero.

### **User-Defined Exceptions**

O/P

→

Balance is too low.

### **Named System Exceptions**

O/P

→

No employee found with that ID.

### **Unhandled (OTHERS) Exception**

O/P

→

Some unexpected error occurred: ORA-00942: table or view does not exist

## **5)PL/SQL – CURSOR**

### **Queries:**

Q1 → select \* from emp;

### **Implicit Cursor**

Q2 →

BEGIN

UPDATE emp SET salary = salary + 1000 WHERE eid = 103;

IF SQL%ROWCOUNT = 1 THEN

DBMS\_OUTPUT.PUT\_LINE('1 row updated successfully.');

END IF;

END;

Q3 → select \* from emp;

### **Explicit Cursor**

Q4 →

DECLARE

CURSOR emp\_cursor IS SELECT eid, name FROM emp;

v\_id emp.eid%TYPE;

v\_name emp.name%TYPE;

BEGIN

OPEN emp\_cursor;

LOOP

FETCH emp\_cursor INTO v\_id, v\_name;

EXIT WHEN emp\_cursor%NOTFOUND;

DBMS\_OUTPUT.PUT\_LINE('ID: ' || v\_id || ', Name: ' || v\_name);

```
END LOOP;  
CLOSE emp_cursor;  
END;
```

### **Cursor FOR Loop**

Q5 →

```
BEGIN  
FOR emp_rec IN (SELECT eid, name FROM emp) LOOP  
DBMS_OUTPUT.PUT_LINE('ID: ' || emp_rec.eid || ', Name: ' || emp_rec.name);  
END LOOP;  
END;
```

### **Parameterized Cursor**

Q6 →

```
DECLARE  
  
CURSOR emp_cursor(p_dept_id NUMBER) IS  
SELECT eid, name FROM emp WHERE dept_id = p_dept_id;  
v_id emp.eid%TYPE;  
v_name emp.name%TYPE;  
  
BEGIN  
OPEN emp_cursor(123); -- passing department id  
LOOP  
FETCH emp_cursor INTO v_id, v_name;  
EXIT WHEN emp_cursor%NOTFOUND;  
DBMS_OUTPUT.PUT_LINE('ID: ' || v_id || ', Name: ' || v_name);  
END LOOP;  
CLOSE emp_cursor;  
END;
```

## **OUTPUT:**

O/P

→

| EID | NAME   | DOJ       | SALARY | DEPT_ID |
|-----|--------|-----------|--------|---------|
| 103 | Cholan | 12-FEB-20 | 30000  | 123     |
| 104 | Hari   | 22-FEB-24 | 20000  | 122     |
| 101 | Arun   | 12-FEB-20 | 40000  | 123     |
| 102 | Bala   | 01-DEC-19 | 50000  | 121     |
| 105 | Guru   | 10-JAN-18 | 60000  | 122     |
| 106 | Mathi  | 01-JUN-20 | 46000  | 121     |

## **Implicit Cursor**

O/P →

1 row updated successfully.

1 row(s) updated.

O/P →

| EID | NAME   | DOJ       | SALARY | DEPT_ID |
|-----|--------|-----------|--------|---------|
| 103 | Cholan | 12-FEB-20 | 31000  | 123     |
| 104 | Hari   | 22-FEB-24 | 20000  | 122     |
| 101 | Arun   | 12-FEB-20 | 40000  | 123     |
| 102 | Bala   | 01-DEC-19 | 50000  | 121     |
| 105 | Guru   | 10-JAN-18 | 60000  | 122     |
| 106 | Mathi  | 01-JUN-20 | 46000  | 121     |

### **Explicit Cursor**

O/P

→

ID: 103, Name: Cholan

ID: 104, Name: Hari

ID: 101, Name: Arun

ID: 102, Name: Bala

ID: 105, Name: Guru

ID: 106, Name: Mathi

### **Cursor FOR Loop**

O/P

→

ID: 103, Name: Cholan

ID: 104, Name: Hari

ID: 101, Name: Arun

ID: 102, Name: Bala

ID: 105, Name: Guru

ID: 106, Name: Mathi

### **Parameterized Cursor**

O/P

→

ID: 103, Name: Cholan

ID: 101, Name: Arun

## **6)PL/SQL – TRIGGER**

### **PL/SQL Queries:**

Q1 → CREATE TABLE students (roll\_no NUMBER PRIMARY KEY,  
name VARCHAR2(50), marks NUMBER);

### **BEFORE INSERT TRIGGER**

Q2 →

```
CREATE OR REPLACE TRIGGER trg_before_insert_students
BEFORE INSERT ON students
FOR EACH ROW
BEGIN
-- If marks not given, set to 0
IF :NEW.marks IS NULL THEN
:NEW.marks := 0;
END IF;
END;
```

### **AFTER INSERT TRIGGER**

Q3 →

```
CREATE OR REPLACE TRIGGER trg_after_insert_students
AFTER INSERT ON students
BEGIN
DBMS_OUTPUT.PUT_LINE('A new student was added!');
END;
```

Q4 → INSERT INTO students (roll\_no, name) VALUES (1, 'John');

Q5 → select \* from students;

### **BEFORE UPDATE TRIGGER**

Q6 →

```
CREATE OR REPLACE TRIGGER trg_before_update_students
BEFORE UPDATE ON students
FOR EACH ROW
BEGIN
    -- Prevent marks from being more than 100
    IF :NEW.marks > 100 THEN
        :NEW.marks := 100;
    END IF;
END;
```

Q7 → UPDATE students SET marks = 120 WHERE roll\_no = 1;

Q8 → select \* from students;

### **AFTER DELETE TRIGGER**

Q9 →

```
CREATE OR REPLACE TRIGGER trg_after_delete_students
AFTER DELETE ON students
BEGIN
    DBMS_OUTPUT.PUT_LINE('A student record was deleted!');
END;
```

Q10 → DELETE FROM students WHERE roll\_no = 1;

### **OUTPUT:**

O/P →

Table Created.

### **BEFORE INSERT TRIGGER**

O/P →

Trigger created.

### **AFTER INSERT TRIGGER**

O/P →

Trigger created.

O/P →

A new student was added!

1 row(s) inserted.

O/P

→

| ROLL_NO | NAME | MARKS |
|---------|------|-------|
| 1       | John | 0     |

### **BEFORE UPDATE TRIGGER**

O/P →

Trigger created.

O/P →

1 row(s) updated.

O/P

→

| ROLL_NO | NAME | MARKS |
|---------|------|-------|
| 1       | John | 100   |

### **AFTER DELETE TRIGGER**

O/P → Trigger created.

O/P → A student record was deleted!

1 row(s) deleted.

## **7)PL/SQL – PACKAGES**

### **PL/SQL Queries:**

#### **Package Specification**

Q1 →

```
CREATE OR REPLACE PACKAGE math_pkg AS  
    FUNCTION square_num(p_num NUMBER) RETURN NUMBER;  
END math_pkg;
```

#### **Package Body**

Q2 →

```
CREATE OR REPLACE PACKAGE BODY math_pkg AS  
    FUNCTION square_num(p_num NUMBER) RETURN NUMBER IS  
    BEGIN  
        RETURN p_num * p_num;  
    END;  
END math_pkg;
```

#### **Test the Package**

Q3 →

```
BEGIN  
    DBMS_OUTPUT.PUT_LINE('Square of 5: ' || math_pkg.square_num(5));  
END;
```

#### **Predefined Package**

Q4 →

```
BEGIN  
    -- Random number between 1 and 50  
    DBMS_OUTPUT.PUT_LINE ('Random Number: ' || ROUND(DBMS_RANDOM.VALUE(1, 50)));  
  
    -- Random uppercase string of length 8  
    DBMS_OUTPUT.PUT_LINE('Random String: ' || DBMS_RANDOM.STRING('U', 8));  
END;
```

## **OUTPUT:**

### **Package Specification**

O/P →

Package created.

### **Package Body**

O/P →

Package Body created.

### **Test the Package**

O/P →

Square of 5: 25

### **Predefined Package**

O/P

→

Random Number: 34

Random String: ZOWXBLHK

## **8)LIBRARY MANAGEMENT SYSTEM**

### **Coding:**

#### **SQL:**

```
create table book ( bookid number primary key,  
    bookname varchar2(30) ,  
    author varchar2(20) ,  
    quan number  
);
```

#### **Form1 (VB 2010):**

Imports System.Data.OleDb

Public Class Form1

' Form-level connection to Oracle 10g

Dim cn As New OleDbConnection("Provider=MSDAORA;Data Source=XE;User  
ID=system;Password=system;")

' --- 1. ADD BOOK ---

Private Sub Button1\_Click(ByVal sender As Object, ByVal e As EventArgs) Handles Button1.Click

' 1. Check if any of your textboxes are empty first

If TextBox1.Text = "" Or TextBox2.Text = "" Or TextBox3.Text = "" Or TextBox4.Text = "" Then

MsgBox("Please fill in all fields before adding a book.")

Exit Sub ' Stops the code here so Oracle doesn't crash

End If

Try

cn.Open()

Dim sql As String = "INSERT INTO book VALUES (" & TextBox1.Text & ", " & TextBox2.Text  
& ", " & TextBox3.Text & ", " & TextBox4.Text & ")"

Dim cmd As New OleDbCommand(sql, cn)

cmd.ExecuteNonQuery()

MsgBox("Book Added Successfully!")

TextBox1.Clear()

TextBox2.Clear()

TextBox3.Clear()

TextBox4.Clear()

Catch ex As Exception

MsgBox("Error adding book: " & ex.Message)

Finally

cn.Close()

End Try

End Sub

' --- 2. ISSUE BOOK ---

Private Sub Button2\_Click(ByVal sender As Object, ByVal e As EventArgs) Handles Button2.Click

' We use InputBox here so you don't have to fill out the textboxes just to issue a book

Dim bid As String = InputBox("Enter the Book ID to Issue:")

If bid = "" Then Exit Sub

Try

cn.Open()

Dim sql As String = "UPDATE book SET quan = quan - 1 WHERE bookid = " & bid

Dim cmd As New OleDbCommand(sql, cn)

Dim rowsAffected As Integer = cmd.ExecuteNonQuery()

If rowsAffected > 0 Then

MsgBox("Book Issued Successfully!")

Else

MsgBox("No record found with ID: " & bid)

End If

Catch ex As Exception

MsgBox("Error issuing book: " & ex.Message)

Finally

cn.Close()

End Try

End Sub

' --- 3. RETURN BOOK ---

Private Sub Button3\_Click(ByVal sender As Object, ByVal e As EventArgs) Handles Button3.Click

Dim bid As String = InputBox("Enter the Book ID to Return:")

If bid = "" Then Exit Sub

Try

cn.Open()

Dim sql As String = "UPDATE book SET quan = quan + 1 WHERE bookid = " & bid

Dim cmd As New OleDbCommand(sql, cn)

Dim rowsAffected As Integer = cmd.ExecuteNonQuery()

If rowsAffected > 0 Then

MsgBox("Book Returned Successfully!")

Else

MsgBox("No record found with ID: " & bid)

End If

Catch ex As Exception

MsgBox("Error returning book: " & ex.Message)

Finally

cn.Close()

End Try

End Sub

' --- 4. VIEW REPORT (Updates the DataGridView at the bottom) ---

Private Sub Button4\_Click(ByVal sender As Object, ByVal e As EventArgs) Handles Button4.Click

Try

Dim sql As String = "SELECT \* FROM book"

' DataAdapter grabs the data without needing to manually open/close the connection

Dim da As New OleDbDataAdapter(sql, cn)

Dim dt As New DataTable()

da.Fill(dt)

DataGridView1.DataSource = dt

Catch ex As Exception

MsgBox("Error loading data: " & ex.Message)

End Try

End Sub

End Class

## Output:

test\_main

lib management

bookID  author

book name  quan

|   | BOOKID | BOOKNAME | AUTHOR  | QUAN |
|---|--------|----------|---------|------|
| ▶ | 101    | RDBMS    | Severus | 55   |
| * |        |          |         |      |

test\_main

lib management

bookID  author

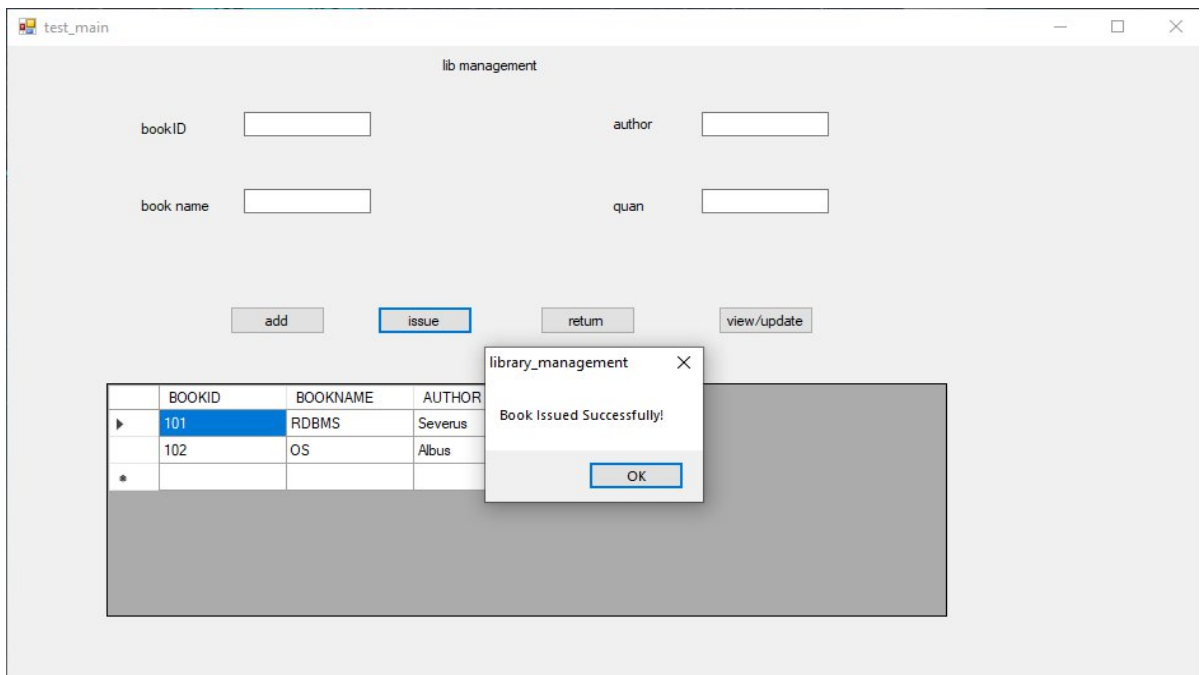
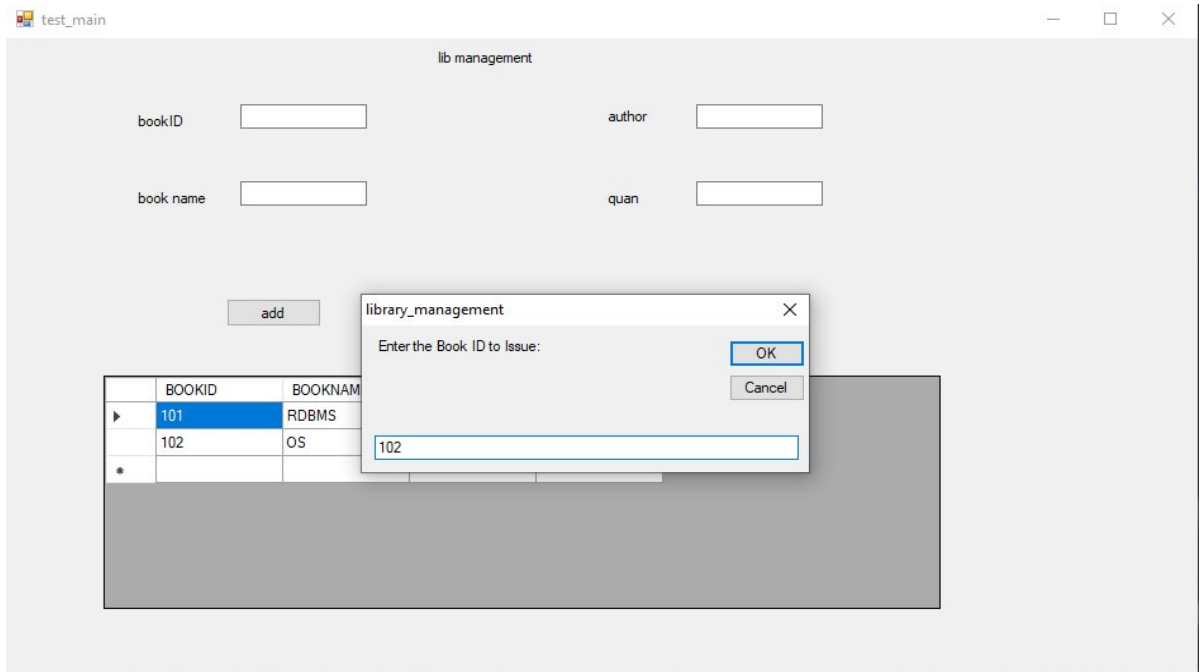
book name  quan

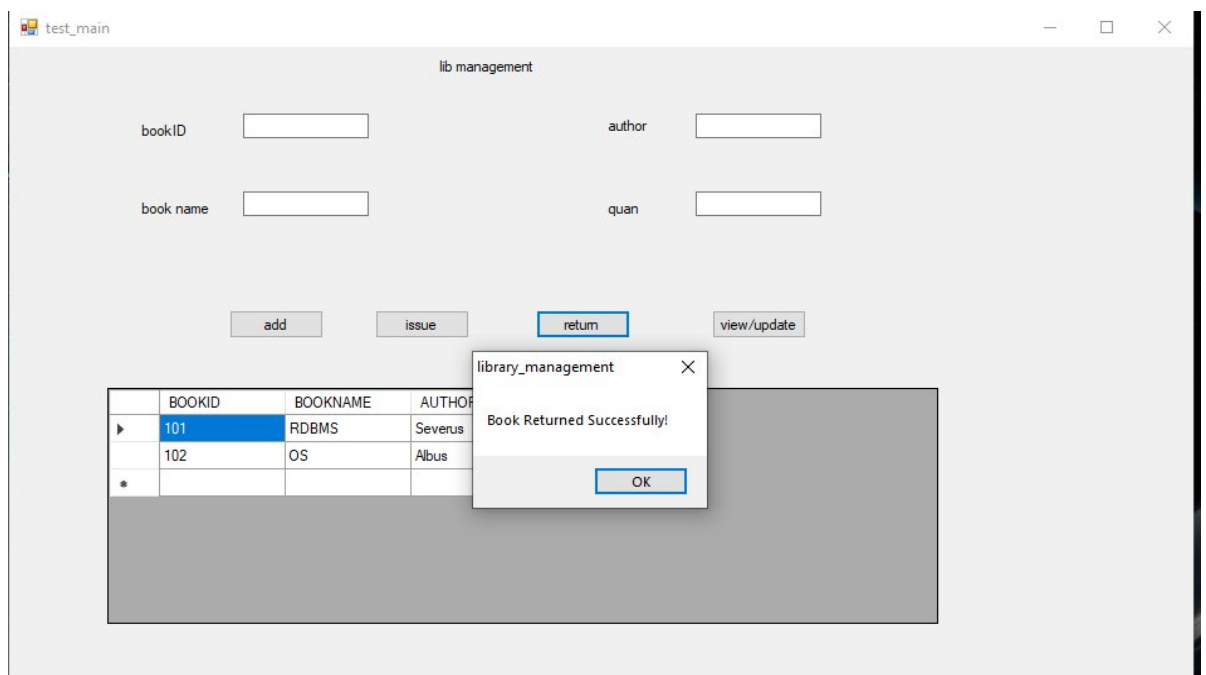
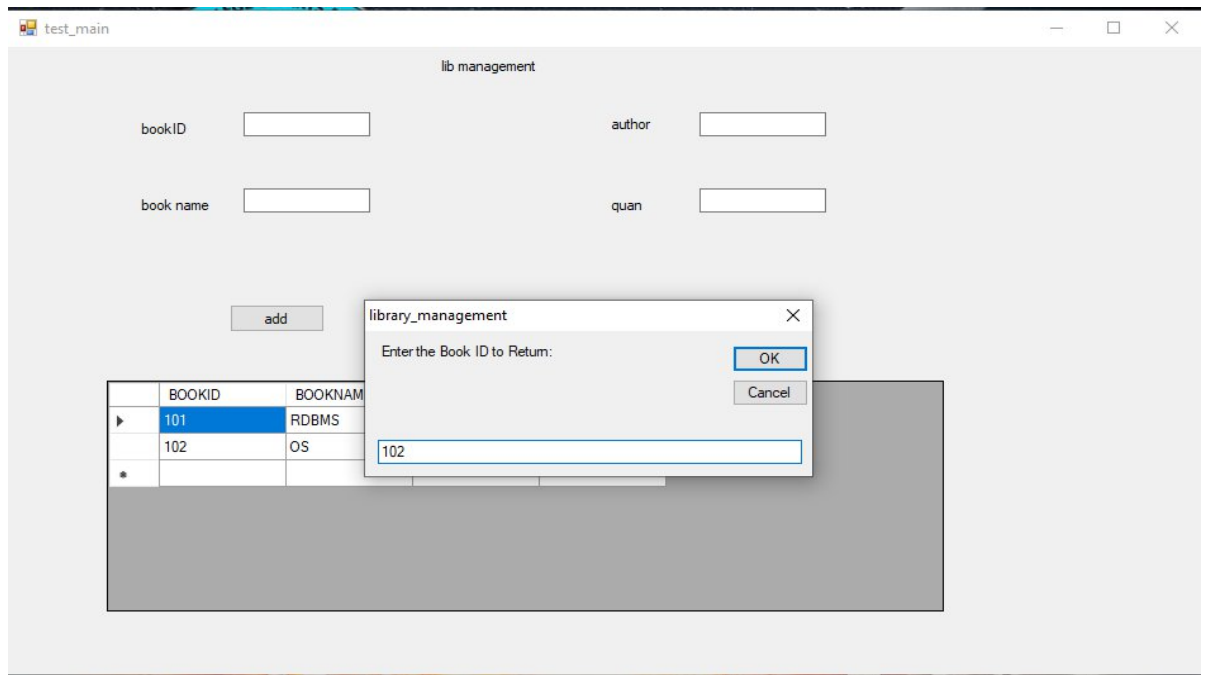
|   | BOOKID | BOOKNAME | AUTHOR  |
|---|--------|----------|---------|
| ▶ | 101    | RDBMS    | Severus |
| * |        |          |         |

library\_management

Book Added Successfully!

OK





## **9)STUDENT MARKSHEET PROCESSING**

### **Coding:**

#### **SQL:**

```
CREATE TABLE st (  
    stid NUMBER PRIMARY KEY,  
    stname VARCHAR2(50),  
    os NUMBER,  
    rdbms NUMBER,  
    cn NUMBER,  
    se NUMBER,  
    total NUMBER,  
    result VARCHAR2(10),  
    percent NUMBER  
);
```

#### **Form1 :**

```
Imports System.Data.OleDb
```

```
Public Class Form1
```

```
    Dim cn As New OleDbConnection("Provider=MSDAORA;Data Source=XE;User  
ID=system;Password=system;")
```

```
    ' --- CALCULATE ---
```

```
    Private Sub Button1_Click(sender As Object, e As EventArgs) Handles Button1.Click  
        TextBox7.Text = (Val(TextBox3.Text) + Val(TextBox4.Text) + Val(TextBox5.Text) +  
Val(TextBox6.Text)).ToString()
```

```
        If Val(TextBox3.Text) <= 30 Or Val(TextBox4.Text) <= 30 Or Val(TextBox5.Text) <= 30  
Or Val(TextBox6.Text) <= 30 Then
```

```
            TextBox8.Text = "FAIL"
```

```
        Else
```

```
            TextBox8.Text = "PASS"
```

```
        End If
```

```
        TextBox9.Text = (Val(TextBox7.Text) / 4).ToString()
```

```
    End Sub
```

' --- INSERT ---

Private Sub Button2\_Click(sender As Object, e As EventArgs) Handles Button2.Click

Button1.PerformClick() ' auto-calculate before insert

Try

cn.Open()

Dim sql As String = "INSERT INTO st VALUES (" & TextBox1.Text & ", " & TextBox2.Text & ", " & TextBox3.Text & ", " & TextBox4.Text & ", " & TextBox5.Text & ", " & TextBox6.Text & ", " & TextBox7.Text & ", " & TextBox8.Text & ", " & TextBox9.Text & ")"

Dim cmd As New OleDbCommand(sql, cn)

cmd.ExecuteNonQuery()

MsgBox("Successfully Inserted")

TextBox1.Clear(): TextBox2.Clear(): TextBox3.Clear()

TextBox4.Clear(): TextBox5.Clear(): TextBox6.Clear()

TextBox7.Clear(): TextBox8.Clear(): TextBox9.Clear()

Catch ex As Exception

MsgBox(ex.Message)

Finally

cn.Close()

End Try

End Sub

' --- DELETE ---

Private Sub Button3\_Click(sender As Object, e As EventArgs) Handles Button3.Click

Dim studentID As String = InputBox("Enter the Student ID to delete:")

If studentID = "" Then Exit Sub

Try

cn.Open()

Dim cmd As New OleDbCommand("DELETE FROM st WHERE stid = " & studentID, cn)

Dim rowsAffected As Integer = cmd.ExecuteNonQuery()

If rowsAffected > 0 Then

MsgBox("Record with ID " & studentID & " deleted successfully!")

Else

MsgBox("No record found with ID: " & studentID)

End If

```
    Catch ex As Exception
        MsgBox(ex.Message)
    Finally
        cn.Close()
    End Try
End Sub
```

```
' --- VIEW ---
```

```
Private Sub Button4_Click(sender As Object, e As EventArgs) Handles Button4.Click
    Try
        Dim da As New OleDbDataAdapter("SELECT * FROM st", cn)
        Dim dt As New DataTable()
        da.Fill(dt)
        DataGridView1.DataSource = dt
    Catch ex As Exception
        MsgBox(ex.Message)
    End Try
End Sub
End Class
```

## Output:

Form1

**STUDENT MARKSHEET PROCESSING**

|              |       |                  |      |
|--------------|-------|------------------|------|
| STUDENT ID   | 12345 | <b>CALCULATE</b> |      |
| STUDENT NAME | Madhu | TOTAL MARKS      | 324  |
| OS           | 67    | RESULT           | PASS |
| RDBMS        | 89    | PERCENTAGE       | 81   |
| CN           | 90    |                  |      |
| SE           | 78    |                  |      |

**REPORT** **INSERT** **DELETE**

student\_db  
Successfully Inserted  
OK

Windows taskbar: Search, 19:25, 26-09-2025

Form1

**STUDENT MARKSHEET PROCESSING**

|              |  |                  |  |
|--------------|--|------------------|--|
| STUDENT ID   |  | <b>CALCULATE</b> |  |
| STUDENT NAME |  | TOTAL MARKS      |  |
| OS           |  | RESULT           |  |
| RDBMS        |  | PERCENTAGE       |  |
| CN           |  |                  |  |
| SE           |  |                  |  |

**REPORT** **INSERT** **DELETE**

student\_db  
Enter the Student ID to delete:  
234  
OK Cancel

Windows taskbar: Rainy days ahead 29°C, Search, 19:40, 26-09-2025

Form1

### STUDENT MARKSHEET PROCESSING

|              |                      |                                          |                      |
|--------------|----------------------|------------------------------------------|----------------------|
| STUDENT ID   | <input type="text"/> | <input type="button" value="CALCULATE"/> |                      |
| STUDENT NAME | <input type="text"/> | TOTAL MARKS                              | <input type="text"/> |
| OS           | <input type="text"/> | RESULT                                   | <input type="text"/> |
| RDBMS        | <input type="text"/> | PERCENTAGE                               | <input type="text"/> |
| CN           | <input type="text"/> |                                          |                      |
| SE           | <input type="text"/> |                                          |                      |

student\_db

Record with ID 234 deleted successfully!

OK

Rainy days ahead  
25°C

Search

ENG IN

19:41  
26-09-2025

DataReport1

Zoom: 100%

| STUDENT ID | stname: | result: | percent: |
|------------|---------|---------|----------|
| 67         | dgh     | PASS    | 69.75    |
| 34         | dpdg    | PASS    | 56.75    |
| 4545       | fpag    | PASS    | 69.75    |

Pages: 1/1

Rainy days ahead  
25°C

Search

ENG IN

19:41  
26-09-2025

## **10) PAYROLL PROCESSING**

### **Coding:**

#### **SQL:**

```
CREATE TABLE emp (  
    empid NUMBER PRIMARY KEY,  
    empname VARCHAR2(50),  
    salary NUMBER,  
    gross NUMBER,  
    net NUMBER );
```

#### **Form1 :**

```
Imports System.Data.OleDb
```

```
Public Class Form1
```

```
    Dim cn As New OleDbConnection("Provider=MSDAORA;Data Source=XE;User ID=system;  
Password=system;") `if data source error occurs use orcl instead of XE
```

```
    ' --- CALCULATE ---
```

```
    Private Sub Button1_Click(sender As Object, e As EventArgs) Handles Button1.Click
```

```
        TextBox4.Text = (Val(TextBox3.Text) * 0.25).ToString()           ' Allowance
```

```
        TextBox5.Text = (Val(TextBox3.Text) * 0.12).ToString()           ' Deduction
```

```
        TextBox6.Text = (Val(TextBox3.Text) + Val(TextBox4.Text)).ToString() ' Gross
```

```
        TextBox7.Text = (Val(TextBox6.Text) - Val(TextBox5.Text)).ToString() ' Net
```

```
    End Sub
```

```
    ' --- INSERT ---
```

```
    Private Sub Button2_Click(sender As Object, e As EventArgs) Handles Button2.Click
```

```
        Button1.PerformClick() ' auto-calculate before insert
```

```
        Try
```

```
            cn.Open()
```

```
            Dim sql As String = "INSERT INTO emp VALUES (" & TextBox1.Text & ", " &  
TextBox2.Text & ", " & TextBox3.Text & ", " & TextBox6.Text & ", " & TextBox7.Text & ")"
```

```
            Dim cmd As New OleDbCommand(sql, cn)
```

```
            cmd.ExecuteNonQuery()
```

```
            MsgBox("Successfully Inserted")
```

```
            TextBox1.Clear(): TextBox2.Clear(): TextBox3.Clear()
```

```
            TextBox4.Clear(): TextBox5.Clear(): TextBox6.Clear()
```

```
            TextBox7.Clear()
```

```
        Catch ex As Exception
```

```
            MsgBox(ex.Message)
```

```
    Finally
```

```
        cn.Close()
```

```
    End Try
```

```
End Sub
```

' --- DELETE ---

Private Sub Button3\_Click(sender As Object, e As EventArgs) Handles Button3.Click

Dim eid As String = InputBox("Enter the EMPLOYEE ID to delete:")

If eid = "" Then Exit Sub

Try

cn.Open()

Dim cmd As New OleDbCommand("DELETE FROM emp WHERE empid = " & eid, cn)

Dim rowsAffected As Integer = cmd.ExecuteNonQuery()

If rowsAffected > 0 Then

MsgBox("Record with ID " & eid & " deleted successfully!")

Else

MsgBox("No record found with ID: " & eid)

End If

Catch ex As Exception

MsgBox(ex.Message)

Finally

cn.Close()

End Try

End Sub

' --- VIEW ---

Private Sub Button4\_Click(sender As Object, e As EventArgs) Handles Button4.Click

Try

Dim da As New OleDbDataAdapter("SELECT \* FROM emp", cn)

Dim dt As New DataTable()

da.Fill(dt)

DataGridView1.DataSource = dt

Catch ex As Exception

MsgBox(ex.Message)

End Try

End Sub

' --- EXIT ---

Private Sub Button5\_Click(sender As Object, e As EventArgs) Handles Button5.Click

Application.Exit()

End Sub

End Class

## Output:

Form1

### EMPLOYEE PAYROLL SYSTEM

|                   |        |
|-------------------|--------|
| EMPLOYEE ID       | 1001   |
| EMPLOYEE NAME     | Anudha |
| SALARY            | 90000  |
| <b>CALCULATE</b>  |        |
| TOTAL ALLOWANCE   | 20000  |
| TOTAL DEDUCTION   | 9600   |
| GROSS SALARY      | 100000 |
| <i>NET SALARY</i> | 90400  |

**INSERT**  
**DELETE**  
**DATA REPORT**  
**EXIT**

PAYROLL  
Successfully Inserted  
OK

Windows taskbar: Search, 19:34, 19-09-2025

Form1

### EMPLOYEE PAYROLL SYSTEM

|                   |  |
|-------------------|--|
| EMPLOYEE ID       |  |
| EMPLOYEE NAME     |  |
| SALARY            |  |
| <b>CALCULATE</b>  |  |
| TOTAL ALLOWANCE   |  |
| TOTAL DEDUCTION   |  |
| GROSS SALARY      |  |
| <i>NET SALARY</i> |  |

**INSERT**  
**DELETE**  
**DATA REPORT**  
**EXIT**

PAYROLL  
Enter the EMPLOYEE ID to delete:  
1234  
OK  
Cancel

Windows taskbar: Search, 19:34, 19-09-2025

Form1

## EMPLOYEE PAYROLL SYSTEM

EMPLOYEE ID

EMPLOYEE NAME

SALARY

**CALCULATE**

TOTAL ALLOWANCE

TOTAL DEDUCTION

GROSS SALARY

*NET SALARY*

**INSERT**

**DELETE**

**DATA REPORT**

**EXIT**

PAYROLL

Record with ID 1237 deleted successfully!

OK

Windows taskbar: Search, 19:34 19-09-2025

DataReport1

Zoom: 100%

| EMPID | NAME   | SALARY | GROSS SAL | NET SALARY |
|-------|--------|--------|-----------|------------|
| 1234  | HARI   | 80000  | 100000    | 90400      |
| 1235  | ABI    | 78000  | 97500     | 88140      |
| 1001  | Amudha | 80000  | 100000    | 90400      |

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Windows taskbar: Search, 19:35 19-09-2025